

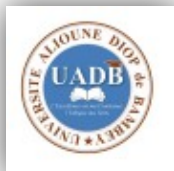
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Un Peuple – Un But – Une Foi

Ministère de l'Enseignement Supérieur et de la Recherche

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UFR Sciences Appliquées et Technologies de l'Information et de la
Communication (**SATIC**)

Département Technologies de l'Information et de la Communication
(**TIC**)

Master professionnelle
Systèmes D'Information (S.I)

TP3 : Application Web dans un conteneur Docker

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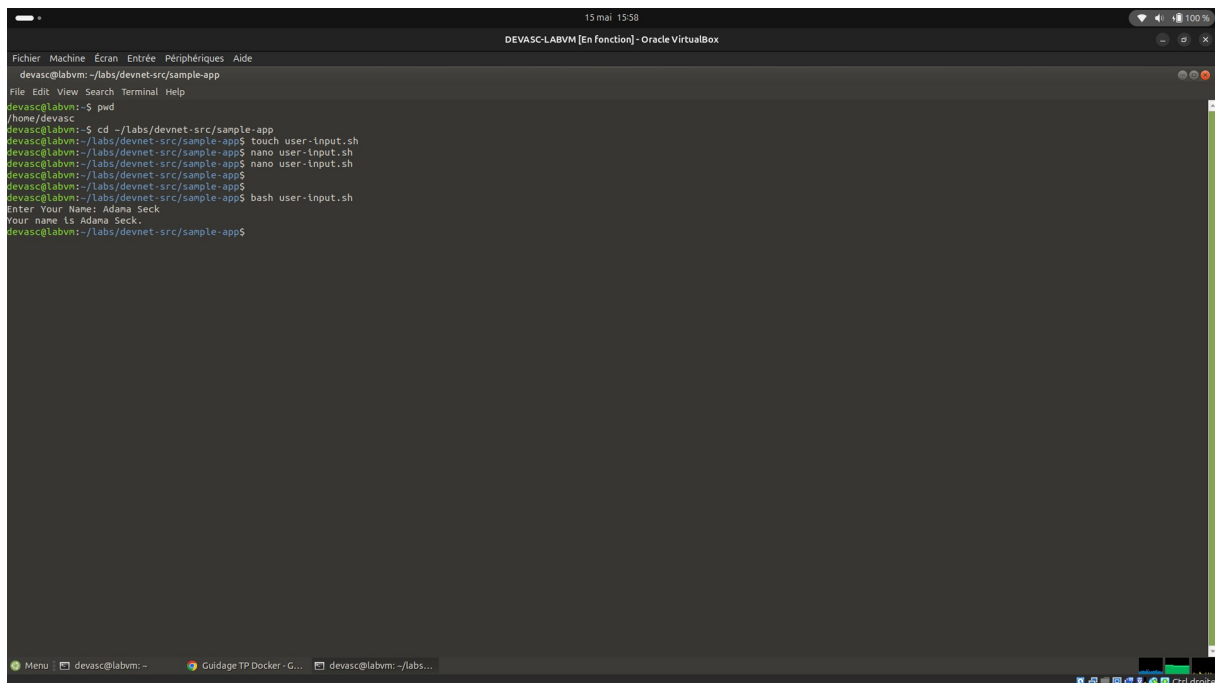
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Introduction à Docker

Docker est une plateforme de conteneurisation qui permet d'empaqueter une application avec toutes ses dépendances dans un environnement isolé et portable. Ce TP explore la création d'une application web conteneurisée à travers des étapes pratiques et progressives.

Partie 1 : Scripts Bash

Les scripts Bash permettent d'automatiser les tâches répétitives dans les environnements Linux. Dans ce TP, nous avons réalisé un script simple qui invite l'utilisateur à saisir son nom et génère une réponse personnalisée en retour.



```
15 mai 15:58
DEVASC-LABVM [En fonction] - Oracle VirtualBox
Fichier Machine Écran Entrée Périphériques Aide
devasc@labvm: ~/labs/devnet-src/sample-app
File Edit View Search Terminal Help
devasc@labvm:~$ pwd
/home/devasc
devasc@labvm:~$ cd ~/labs/devnet-src/sample-app
devasc@labvm:~/labs/devnet-src/sample-app$ touch user-input.sh
devasc@labvm:~/labs/devnet-src/sample-app$ nano user-input.sh
devasc@labvm:~/labs/devnet-src/sample-app$ nano user-input.sh
devasc@labvm:~/labs/devnet-src/sample-app$
devasc@labvm:~/labs/devnet-src/sample-app$
devasc@labvm:~/labs/devnet-src/sample-app$ bash user-input.sh
Enter Your Name: Adana Seck
Your name is Adana Seck.
devasc@labvm:~/labs/devnet-src/sample-app$
```

Figure 1 : Script Bash – Saisie du nom et réponse personnalisée

Partie 2 : Application Flask

Flask est un micro-framework Python léger destiné à la création d'applications web. L'application développée affiche l'adresse IP du client visitant la page, tout en servant dynamiquement les fichiers HTML et CSS.

```
15 mai 16:02
DEVASC-LABVM [En fonction] - Oracle VirtualBox

Fichier Machine Écran Entrée Périphériques Aide
devasc@labvm: ~/labs/devnet-src/sample-app
File Edit View Search Terminal Help

devasc@labvm:~/labs/devnet-src/sample-app$ pip3 install flask
Requirement already satisfied: flask in /home/devasc/.local/lib/python3.8/site-packages (1.1.2)
Requirement already satisfied: itsdangerous<0.24 in /home/devasc/.local/lib/python3.8/site-packages (from flask) (1.1.0)
Requirement already satisfied: Werkzeug<0.15 in /home/devasc/.local/lib/python3.8/site-packages (from flask) (1.0.1)
Requirement already satisfied: click>=5.1 in /home/devasc/.local/lib/python3.8/site-packages (from flask) (7.1.2)
Requirement already satisfied: Jinja2>=2.10.1 in /home/devasc/.local/lib/python3.8/site-packages (from flask) (2.11.2)
Requirement already satisfied: MarkupSafe<0.23 in /home/devasc/.local/lib/python3.8/site-packages (from Jinja2>=2.10.1->flask) (1.1.1)
devasc@labvm:~/labs/devnet-src/sample-app$ nano sample_app.py
devasc@labvm:~/labs/devnet-src/sample-app$
```

Figure 2 : Code de l'application Flask affichant l'adresse IP du client

```
15 mai 16:02
DEVASC-LABVM [En fonction] - Oracle VirtualBox

Fichier Machine Écran Entrée Périphériques Aide
devasc@labvm: ~/labs/devnet-src/sample-app
File Edit View Search Terminal Help

GNU nano 4.8 sample_app.py
from flask import Flask
from flask import request

sample = Flask(__name__)

@sample.route("/")
def main():
    return "You are calling me from " + request.remote_addr + "\n"

if __name__ == "__main__":
    sample.run(host="0.0.0.0", port=8080)
```

Figure 3 : Rendu de l'application Flask dans le navigateur

```
15 mai 16:04
DEVASC-LABVM [En fonction] - Oracle VirtualBox

Fichier Machine Écran Entrée Périphériques Aide
devasc@labvm: ~/labs/devnet-src/sample-app

File Edit View Search Terminal Help

devasc@labvm:~/labs/devnet-src/sample-app$ pip3 install flask
Requirement already satisfied: flask in /home/devasc/.local/lib/python3.8/site-packages (1.1.2)
Requirement already satisfied: itsdangerous>=0.24 in /home/devasc/.local/lib/python3.8/site-packages (from flask) (1.1.0)
Requirement already satisfied: Werkzeug>=0.15 in /home/devasc/.local/lib/python3.8/site-packages (from flask) (1.0.1)
Requirement already satisfied: click>=5.1 in /home/devasc/.local/lib/python3.8/site-packages (from flask) (7.1.2)
Requirement already satisfied: Jinja2>=2.10.1 in /home/devasc/.local/lib/python3.8/site-packages (from flask) (2.11.2)
Requirement already satisfied: MarkupSafe>=0.23 in /home/devasc/.local/lib/python3.8/site-packages (from Jinja2>=2.10.1->flask) (1.1.1)
devasc@labvm:~/labs/devnet-src/sample-app$ nano sample_app.py
devasc@labvm:~/labs/devnet-src/sample-app$ python3 sample_app.py
* Serving Flask app "sample_app" (lazy loading)
* Environment: production
  WARNING: This is a development server. Do not use it in a production deployment.
  Use a production WSGI server instead.
* Debug mode: off
* Running on http://0.0.0.0:8080/ (Press CTRL+C to quit)
127.0.0.1 - - [15/May/2026 16:03:54] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [15/May/2026 16:03:54] "GET /favicon.ico HTTP/1.1" 404 -
```

Figure 4 : Structure et organisation des fichiers du projet Flask

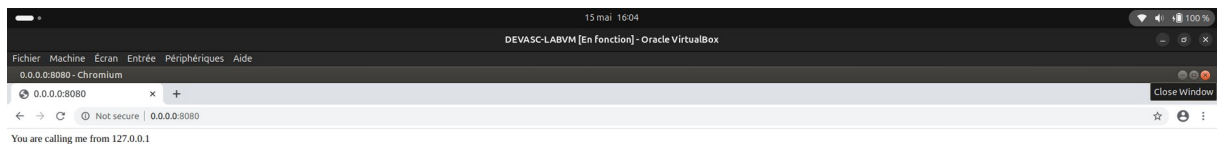
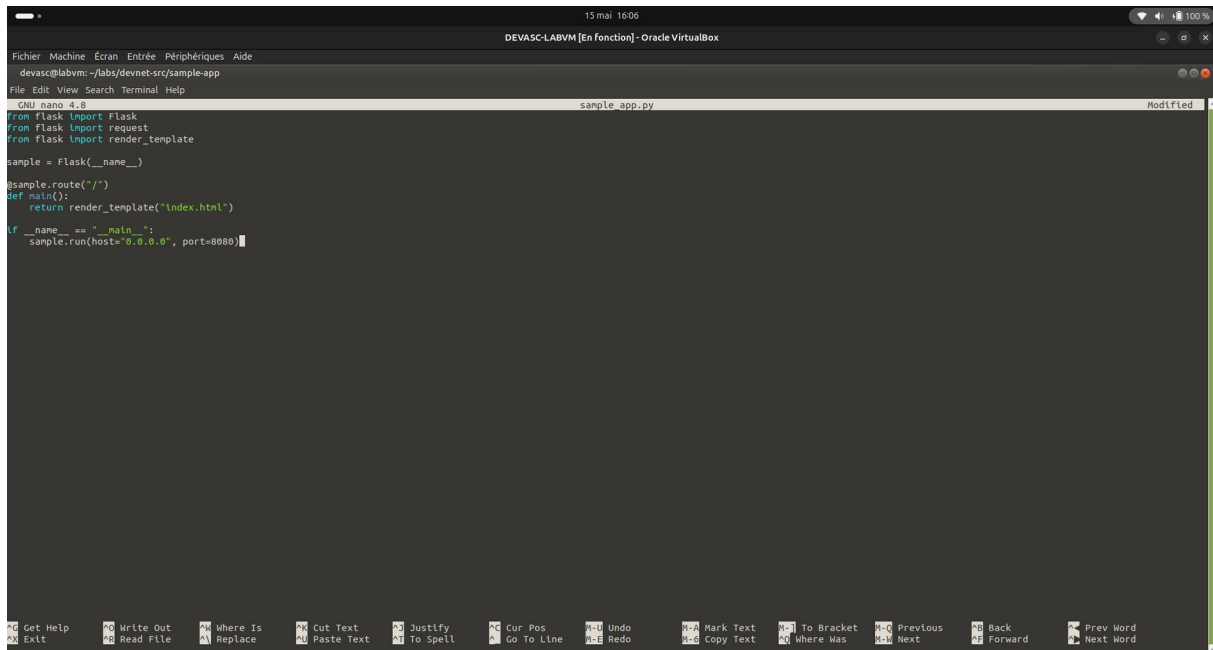


Figure 5 : Contenu du fichier de configuration HTML et CSS de l'application

Partie 3 : Configuration HTML/CSS

L'application a été enrichie d'une interface visuelle grâce à l'intégration de fichiers HTML et CSS. Cette interface permet d'obtenir une page web stylisée, avec un fond bleu acier clair et un titre principal formaté en balise H1.



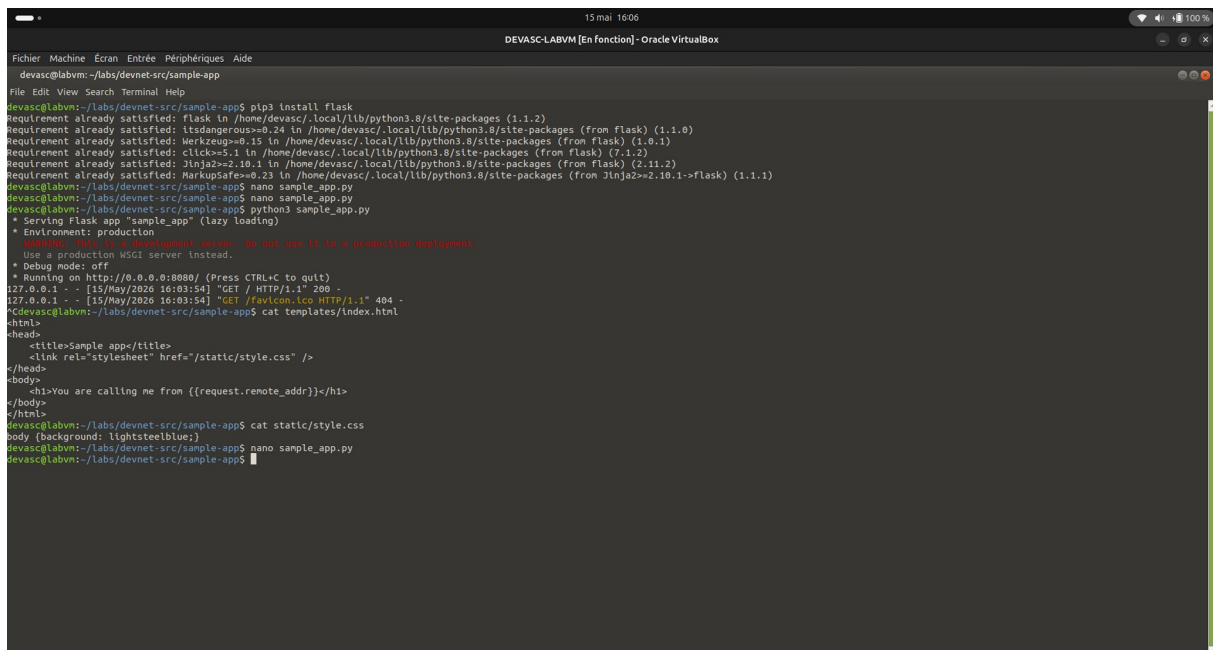
```
from flask import Flask
from flask import request
from flask import render_template

sample = Flask(__name__)

@sample.route("/")
def main():
    return render_template("index.html")

if __name__ == "__main__":
    sample.run(host="0.0.0.0", port=8080)
```

Figure 6 : Interface web stylisée avec fond bleu acier et titre en balise H1



```
devasc@labvm:~/labs/devnet-src/sample-app$ pip3 install flask
Requirement already satisfied: flask in /home/devasc/.local/lib/python3.8/site-packages (1.1.2)
Requirement already satisfied: itsdangerous<=0.24 in /home/devasc/.local/lib/python3.8/site-packages (from flask) (1.1.0)
Requirement already satisfied: Werkzeug<=0.15 in /home/devasc/.local/lib/python3.8/site-packages (from flask) (1.0.1)
Requirement already satisfied: click<=5.1 in /home/devasc/.local/lib/python3.8/site-packages (from flask) (7.1.2)
Requirement already satisfied: Jinja2<=2.10.1 in /home/devasc/.local/lib/python3.8/site-packages (from flask) (2.11.2)
Requirement already satisfied: MarkupSafe<=0.23 in /home/devasc/.local/lib/python3.8/site-packages (from Jinja2<=2.10.1->flask) (1.1.1)
devasc@labvm:~/labs/devnet-src/sample-app$ pip3 run sample_app.py
 * Serving Flask app "sample_app" (lazy loading)
 * Environment: production
   WARNING: This is a development server. Do not use it in a production deployment.
   Use a production WSGI server instead.
 * Debug mode: off
 * Running on http://0.0.0.0:8080/ (Press CTRL+C to quit)
127.0.0.1 - - [15/May/2020 16:03:54] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [15/May/2020 16:03:54] "GET /favicon.ico HTTP/1.1" 404 -
devasc@labvm:~/labs/devnet-src/sample-app$ cat templates/index.html
<html>
<head>
<title>Sample app</title>
<link rel="stylesheet" href="/static/style.css" />
</head>
<body>
<h1>You are calling me from {{request.remote_addr}}</h1>
</body>
</html>
devasc@labvm:~/labs/devnet-src/sample-app$ cat static/style.css
body {background: lightsteelblue;}
devasc@labvm:~/labs/devnet-src/sample-app$ nano sample_app.py
devasc@labvm:~/labs/devnet-src/sample-app$
```

Figure 7 : Aperçu de la page web rendue dans le navigateur

```
15 mai 1607
DEVASC-LABVM [En fonction] - Oracle VirtualBox

Fichier Machine Écran Entrée Périphériques Aide
devasc@labvm:~/labs/devnet-src/sample-app$
File Edit View Search Terminal Help
devasc@labvm:~/labs/devnet-src/sample-app$ pip3 install flask
Requirement already satisfied: flask in /home/devasc/.local/lib/python3.8/site-packages (1.1.2)
Requirement already satisfied: itsdangerous>=0.24 in /home/devasc/.local/lib/python3.8/site-packages (from flask) (1.1.0)
Requirement already satisfied: Werkzeug>=0.15 in /home/devasc/.local/lib/python3.8/site-packages (from flask) (1.0.1)
Requirement already satisfied: click>=5.1 in /home/devasc/.local/lib/python3.8/site-packages (from flask) (7.1.2)
Requirement already satisfied: Jinja2>=2.10.1 in /home/devasc/.local/lib/python3.8/site-packages (from flask) (2.11.2)
Requirement already satisfied: MarkupSafe>=0.23 in /home/devasc/.local/lib/python3.8/site-packages (from Jinja2>=2.10.1->flask) (1.1.1)
devasc@labvm:~/labs/devnet-src/sample-app$ nano sample_app.py
devasc@labvm:~/labs/devnet-src/sample-app$ python3 sample_app.py
* Serving Flask app "sample_app" (lazy loading)
* Environment: production
  WARNING: This is a development server. Do not use it in a production deployment.
  Use a production WSGI server instead.
* Debug mode: off
* Running on http://0.0.0.0:8080/ (Press CTRL+C to quit)
127.0.0.1 - - [15/May/2026 16:03:54] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [15/May/2026 16:03:54] "GET /favicon.ico HTTP/1.1" 404 -
devasc@labvm:~/labs/devnet-src/sample-app$ cat templates/index.html
<html>
<head>
  <title>Sample app</title>
  <link rel="stylesheet" href="/static/style.css" />
</head>
<body>
  <h1>You are calling me from {{request.remote_addr}}</h1>
</body>
</html>
devasc@labvm:~/labs/devnet-src/sample-app$ cat static/style.css
body {background: lightsteelblue;}
devasc@labvm:~/labs/devnet-src/sample-app$ nano sample_app.py
devasc@labvm:~/labs/devnet-src/sample-app$ python3 sample_app.py
* Serving Flask app "sample_app" (lazy loading)
* Environment: production
  WARNING: This is a development server. Do not use it in a production deployment.
  Use a production WSGI server instead.
* Debug mode: off
* Running on http://0.0.0.0:8080/ (Press CTRL+C to quit)
127.0.0.1 - - [15/May/2026 16:06:55] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [15/May/2026 16:06:55] "GET /static/style.css HTTP/1.1" 200 -
```

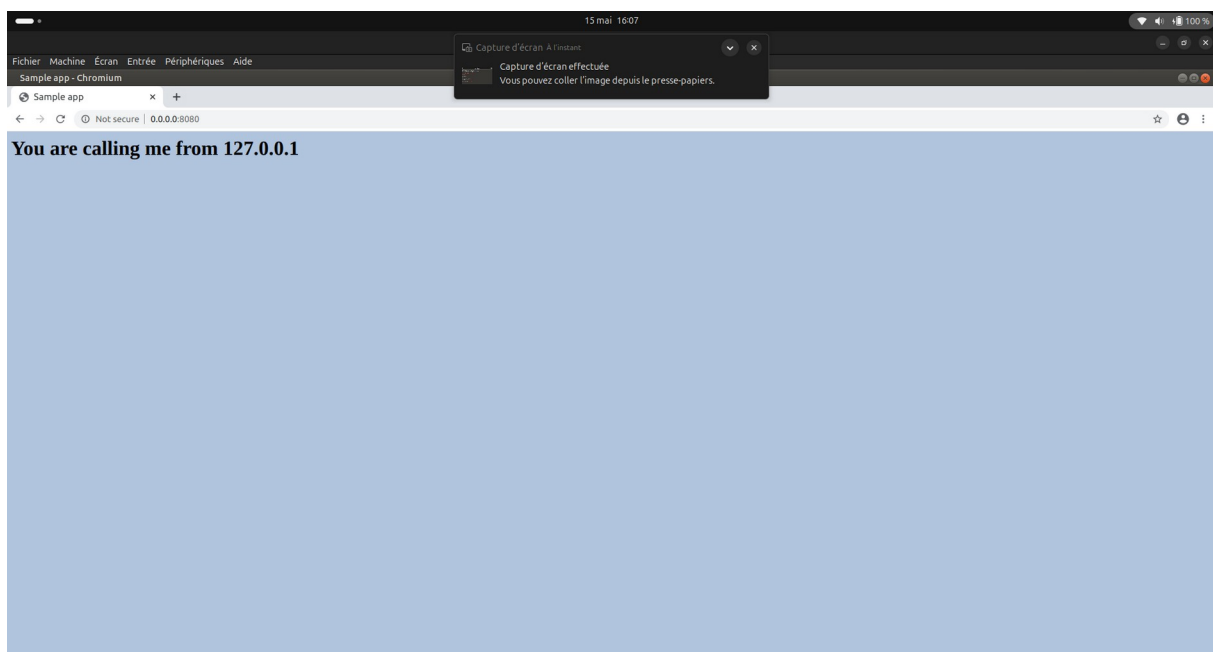


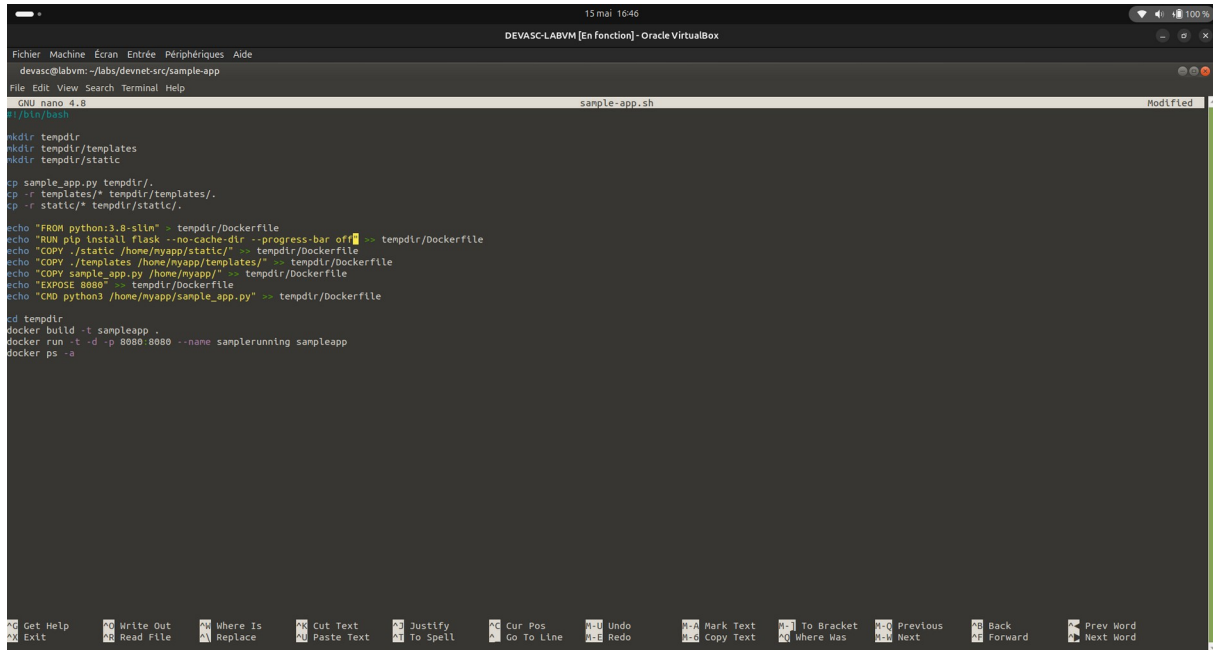
Figure 8 : Script Bash d'automatisation du déploiement du conteneur Docker

Partie 4 : Automatisation Docker

Le script Bash final automatise entièrement le déploiement du conteneur Docker en enchaînant plusieurs étapes : création des répertoires, copie des fichiers sources, génération du Dockerfile, construction de l'image et lancement du conteneur.

- Dockerfile et ses instructions

Le Dockerfile définit les étapes de construction de l'image. Les instructions clés sont : FROM pour spécifier l'image de base, RUN pour exécuter les commandes d'installation, COPY pour intégrer les fichiers, EXPOSE pour déclarer le port d'écoute et CMD pour définir la commande de démarrage.



```
15 mai 16:46
DEVASC-LABVM [En fonction] - Oracle VM VirtualBox

Fichier Machine Écran Entrée Périphériques Aide
devasc@labvm: ~/labs/devnet-src/sample-app
File Edit View Search Terminal Help
GNU nano 4.8 sample-app.sh Modified

#!/bin/bash

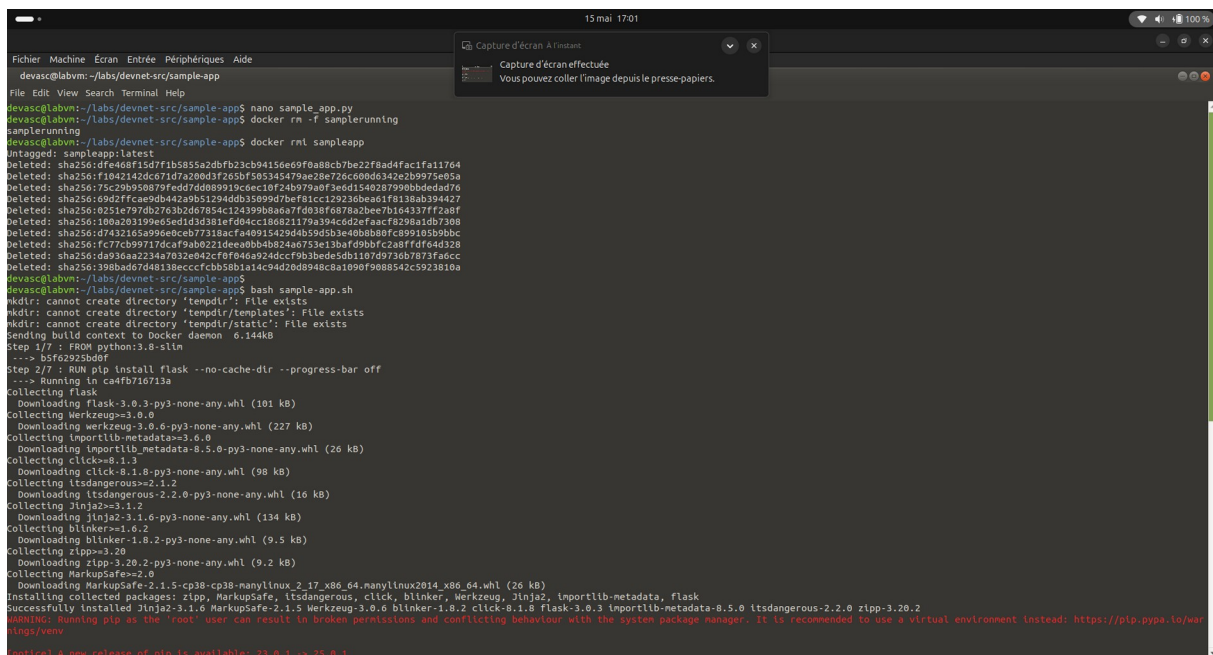
mkdir tmpdir
mkdir tmpdir/templates
mkdir tmpdir/static

cp sample_app.py tmpdir/
cp -r templates/* tmpdir/templates/
cp -r static/* tmpdir/static/

echo "FROM python:3.8-slim" >> tmpdir/Dockerfile
echo "RUN pip install flask --no-cache-dir --progress-bar off" >> tmpdir/Dockerfile
echo "COPY ./static /home/myapp/static/" >> tmpdir/Dockerfile
echo "COPY ./templates /home/myapp/templates/" >> tmpdir/Dockerfile
echo "COPY sample_app.py /home/myapp/" >> tmpdir/Dockerfile
echo "EXPOSE 8080" >> tmpdir/Dockerfile
echo "CMD python3 /home/myapp/sample_app.py" >> tmpdir/Dockerfile

cd tmpdir
docker build -t sampleapp .
docker run -t -d -p 8080:8080 --name samplerunning sampleapp
docker ps -a
```

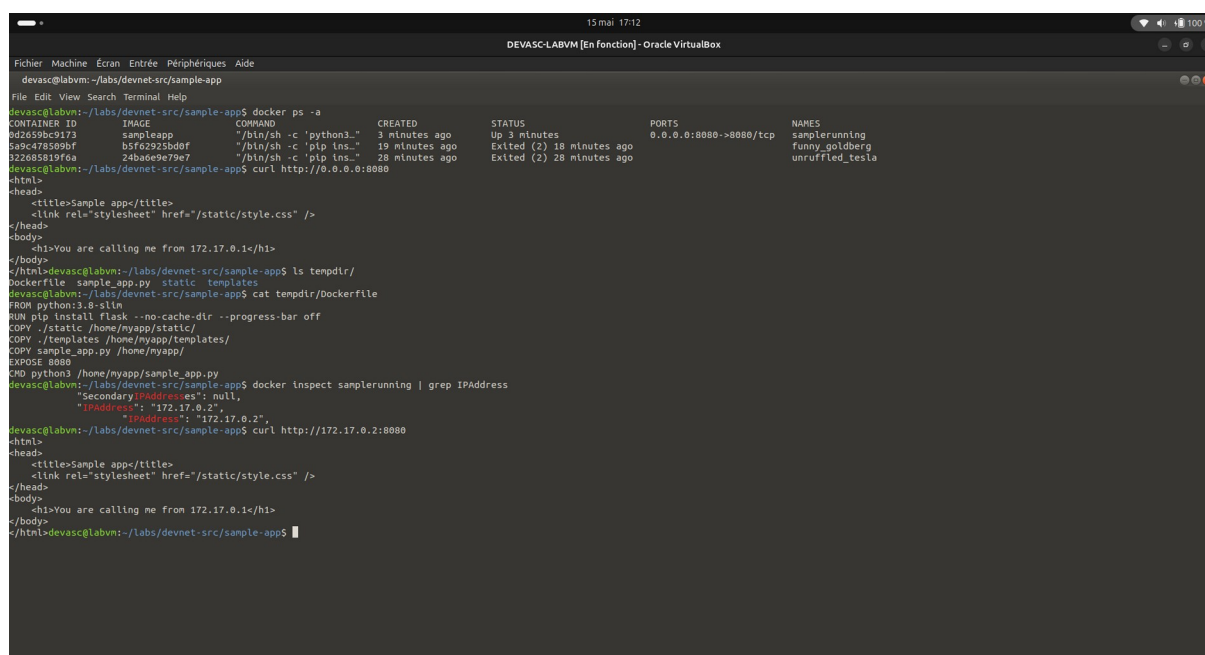
Figure 9 : Contenu du Dockerfile définissant les instructions de construction de l'image



```
15 mai 17:01
Fichier Machine Écran Entrée Périphériques Aide
devasc@labvm: ~/labs/devnet-src/sample-app
File Edit View Search Terminal Help
devasc@labvm:~/labs/devnet-src/sample-app$ nano sample_app.py
devasc@labvm:~/labs/devnet-src/sample-app$ docker rm -f samplerunning
samplerunning
devasc@labvm:~/labs/devnet-src/sample-app$ docker rmi sampleapp
Untagged: sampleapp:latest
Deleted: sha256:dfe408f15d7f1b5855a2dbfb23cb94156e9f0a88cb7be22f8ad4fac1f11764
Deleted: sha256:f1042142dc671d7a20d3f265bf505345479ae28e726c00d6342e2b9975e05a
Deleted: sha256:75c290508797fed7d0809919csec10f24b979a0f3e6d1546287990b0dedad76
Deleted: sha256:69d2ffca08042e9b512e4db3599d70bf91ccc122326ba61f818ab394427
Deleted: sha256:0251e797db2763b2d67854c124399b8a6a7fd938f6878a2bee7b164337f2a8f
Deleted: sha256:100a203199e65ed1d3d381ef04cc180821179a394c6d2efaac6f298a1db7308
Deleted: sha256:d74221650996ebc77318acfa4991542d4b59d5b3e40bb80fc09195b9bbc
Deleted: sha256:fc77cb99717dca9ab0221dea0bb4b824a6753e13baf9bbfc2a8ffdf64d328
Deleted: sha256:da936a2234a7632e042cf0f046a924dc9f9b3bede5db1107d9736b7873fa6cc
Deleted: sha256:39b0a6f04813b0ccfcb5b0b1a14c94d2d08948c8a1090f9808542c5923010a
devasc@labvm:~/labs/devnet-src/sample-app$
devasc@labvm:~/labs/devnet-src/sample-app$ bash sample-app.sh
mkdir: cannot create directory 'tmpdir': File exists
mkdir: cannot create directory 'tmpdir/templates': File exists
mkdir: cannot create directory 'tmpdir/static': File exists
Sending build context to Docker daemon  6.144kB
Step 1/7 : FROM python:3.8-slim
--> b5f62925b09f
Step 2/7 : RUN pip install flask --no-cache-dir --progress-bar off
--> Running in ca4fb716713a
collecting flask
  Downloading flask-3.0.3-py3-none-any.whl (101 kB)
collecting werkzeug
  Downloading werkzeug-3.0.6-py3-none-any.whl (227 kB)
collecting importlib-metadata
  Downloading importlib_metadata-8.5.0-py3-none-any.whl (26 kB)
collecting click
  Downloading click-8.1.8-py3-none-any.whl (98 kB)
collecting itsdangerous
  Downloading itsdangerous-2.2.0-py3-none-any.whl (16 kB)
collecting Jinja2
  Downloading Jinja2-3.1.6-py3-none-any.whl (134 kB)
collecting blinker
  Downloading blinker-1.8.2-py3-none-any.whl (9.5 kB)
collecting zipp
  Downloading zipp-3.20.2-py3-none-any.whl (9.2 kB)
collecting MarkupSafe
  Downloading MarkupSafe-2.1.5-cp38-cp38-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (26 kB)
Installing collected packages: zipp, MarkupSafe, itsdangerous, click, blinker, Werkzeug, Jinja2, importlib-metadata, flask
Successfully installed Jinja2-3.1.6 MarkupSafe-2.1.5 Werkzeug-3.0.6 blinker-1.8.2 click-8.1.8 flask-3.0.3 importlib-metadata-8.5.0 itsdangerous-2.2.0 zipp-3.20.2
WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour with the system package manager. It is recommended to use a virtual environment instead: https://pip.pypa.io/warnings/venv
(Python) A new release of pip is available: 23.0.1 -> 23.0.1
```



Figure 10 : Construction de l'image et lancement du conteneur Docker



```
devasc@labvm:~/labs/devnet-src/sample-app$ docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                    NAMES
bd2659bc9173   sampleapp     "/bin/sh -c 'python3..." 3 minutes ago  Up 3 minutes  0.0.0.0:8080->8080/tcp    samplerunning
5a9c4785b9bf   b5f62925bd0f "/bin/sh -c 'pip ins..." 19 minutes ago Exited (2) 18 minutes ago funny_goldberg
b22b85819f6a   24ba0e9e79e7 "/bin/sh -c 'pip ins..." 28 minutes ago Exited (2) 28 minutes ago unruffled_testa

devasc@labvm:~/labs/devnet-src/sample-app$ curl http://0.0.0.0:8080
<html>
<head>
  <title>Sample app</title>
  <link rel="stylesheet" href="/static/style.css" />
</head>
<body>
  <h1>You are calling me from 172.17.0.1</h1>
</body>
</html>devasc@labvm:~/labs/devnet-src/sample-app$ ls tempdir/
Dockerfile  sample_app.py  static  templates
devasc@labvm:~/labs/devnet-src/sample-app$ cat tempdir/Dockerfile
FROM python:3.8-slim
RUN pip install flask --no-cache-dir --progress-bar off
COPY ./static /home/myapp/static/
COPY ./templates /home/myapp/templates/
COPY sample_app.py /home/myapp/
EXPOSE 8080
CMD python3 /home/myapp/sample_app.py
devasc@labvm:~/labs/devnet-src/sample-app$ docker inspect samplerunning | grep IPAddress
    "SecondaryIPAddresses": null,
    "IPAddress": "172.17.0.2",
    "IPAddress": "172.17.0.2",
devasc@labvm:~/labs/devnet-src/sample-app$ curl http://172.17.0.2:8080
<html>
<head>
  <title>Sample app</title>
  <link rel="stylesheet" href="/static/style.css" />
</head>
<body>
  <h1>You are calling me from 172.17.0.1</h1>
</body>
</html>devasc@labvm:~/labs/devnet-src/sample-app$
```

Figure 11 : Résultat de la commande docker ps – conteneur actif et port exposé

- Vérification et gestion du conteneur

Après le lancement, il est possible de vérifier l'état du conteneur avec `docker ps`, d'accéder à son shell interactif avec `docker exec`, et d'arrêter ou de redémarrer le service à l'aide de `docker stop` et `docker start`.

Conclusion

Ce travail pratique nous a permis de découvrir Docker de manière progressive et concrète. En partant d'un simple script Bash, nous avons construit pas à pas une application web conteneurisée, intégrant Flask, HTML, CSS, et un Dockerfile complet. Chaque étape a renforcé notre compréhension des mécanismes de conteneurisation : isolation de l'environnement, portabilité de l'application et automatisation du déploiement.

Au-delà des aspects techniques, ce TP illustre l'intérêt réel de Docker dans un contexte professionnel : garantir qu'une application fonctionne de la même façon sur n'importe quelle machine, sans se soucier des différences d'environnement. C'est une

compétence aujourd'hui incontournable dans les métiers du développement et de l'administration système.

En somme, cette expérience nous a donné de bonnes bases pour aller plus loin, notamment vers l'orchestration de conteneurs avec des outils comme Docker Compose ou Kubernetes, qui constituent la prochaine étape naturelle dans la maîtrise des technologies cloud et de virtualisation.

